Source Credibility, Belief, and Sharing Behavior

A Causal Machine Learning Analysis

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Contents

1 Introduction

1.1 Research Objective

The increasing spread of "fake news" represents one of the great challenges societies face in the 21st century. The aim of the research is to investigate how sources (fake vs. real) influence whether people believe and share what they read. Specifically, the objectives are: (1) Investigate how imitating well-known and credible sources affects people's belief in the information provided; (2) Investigate how the credibility of these sources affects the likelihood of sharing the news; (3) Explore how factors like age, education, income influence the above dynamics; (4) Uncover CATE and treatment heterogeneity using machine learning techniques blending with conventional econometric methods

1.2 Research Questions and Hypotheses

Bauer (2020) proposes two main hypotheses. First, individuals are more likely to believe news reports from real sources than from fake sources (H1a). Second, individuals are more likely to share news originating from real sources (H1b). While these hypotheses focus on average effects, an equally important question is whether the impact of source credibility varies across individuals. We therefore adopt an exploratory approach to study treatment heterogeneity, without imposing strong prior assumptions.

Demographic factors such as age, education, and income are likely to play a key role in shaping responses to information. Age, in particular, has been shown to influence media consumption and online participation (Hargittai and Walejko, 2008; Loges and Jung, 2001). However, the empirical evidence is mixed: some studies suggest that younger individuals are less influenced by mainstream news sources (American Press Institute, 2017; YouGov, 2017; PricewaterhouseCoopers, 2018), while others find that older individuals exhibit greater skepticism toward media (Gunther, 1992; Metzger et al., 2003). Education and income are also important, as they are associated with the ability to distinguish between credible and non-credible information sources (Fogg et al., 2001). At the same time, there is ongoing debate about whether higher education reduces or reinforces biases in information processing (Kahan, 2013; Tappin, Pennycook, and Rand, 2018).

To test the main hypotheses, we employ t-tests for average treatment effects and linear regression models for interaction effects. In addition, we conduct robustness and manipulation checks to ensure the validity of the experimental design. However, analyzing treatment heterogeneity using conventional methods is challenging due to the large number of potential covariates. In particular, conducting multiple hypothesis tests leads to concerns about statistical validity, as p-values become unreliable under multiple testing (Athey and Imbens, 2017). To address this issue, we complement standard econometric approaches with machine learning methods that allow for a flexible and data-driven exploration of heterogeneous treatment effects.

2 Methodology

2.1 Data Description

We use a dataset 'Fake News' including 1980 individuals with a selected set of variables from the original dataset collected by Bauer (2020) showing how news sources influence people's beliefs and intentions to share news reports. The analysis includes demographic control variables such as `age`, `education`, `income`, and `gender`. `Age` is grouped into five categories, while `education` reflects the highest level attained. `Income` is divided into eleven brackets with 500 euro increments, and gender is coded as a binary variable. The main outcomes of interest as outcome variables are `belief` in the news report and sharing intentions. Sharing behavior is measured across multiple platforms, including `email`, `Facebook`, `Twitter`, and `WhatsApp`.

Table 1: Summary statistics key variables

Variable	N	Mean	St. Dev.	Min	Max
Age	1980	46.53	16.07	18	89
Belief	1979	3.46	1.57	0	6
Education	1962	3.09	1.43	1	5
Gender	1980	0.49	0.50	0	1
Income	1964	5.71	2.70	1	11
Share: Email	1930	0.05	0.23	0	1
Share: Facebook	1210	0.11	0.31	0	1
Share: Twitter	292	0.13	0.33	0	1
Share: WhatsApp	1639	0.09	0.28	0	1
Treatment: Source	1980	0.49	0.50	0	1

The summary statistics from Table 1 suggest that the main variables are broadly balanced in terms of their mean values, although some degree of skewness is evident, as illustrated in Figure 1. In addition, sharing behavior varies across platforms, with Twitter being the least used channel. This pattern may introduce limitations that will be discussed later. Figure 1 presents the distributions of the key demographic variables used in the analysis.

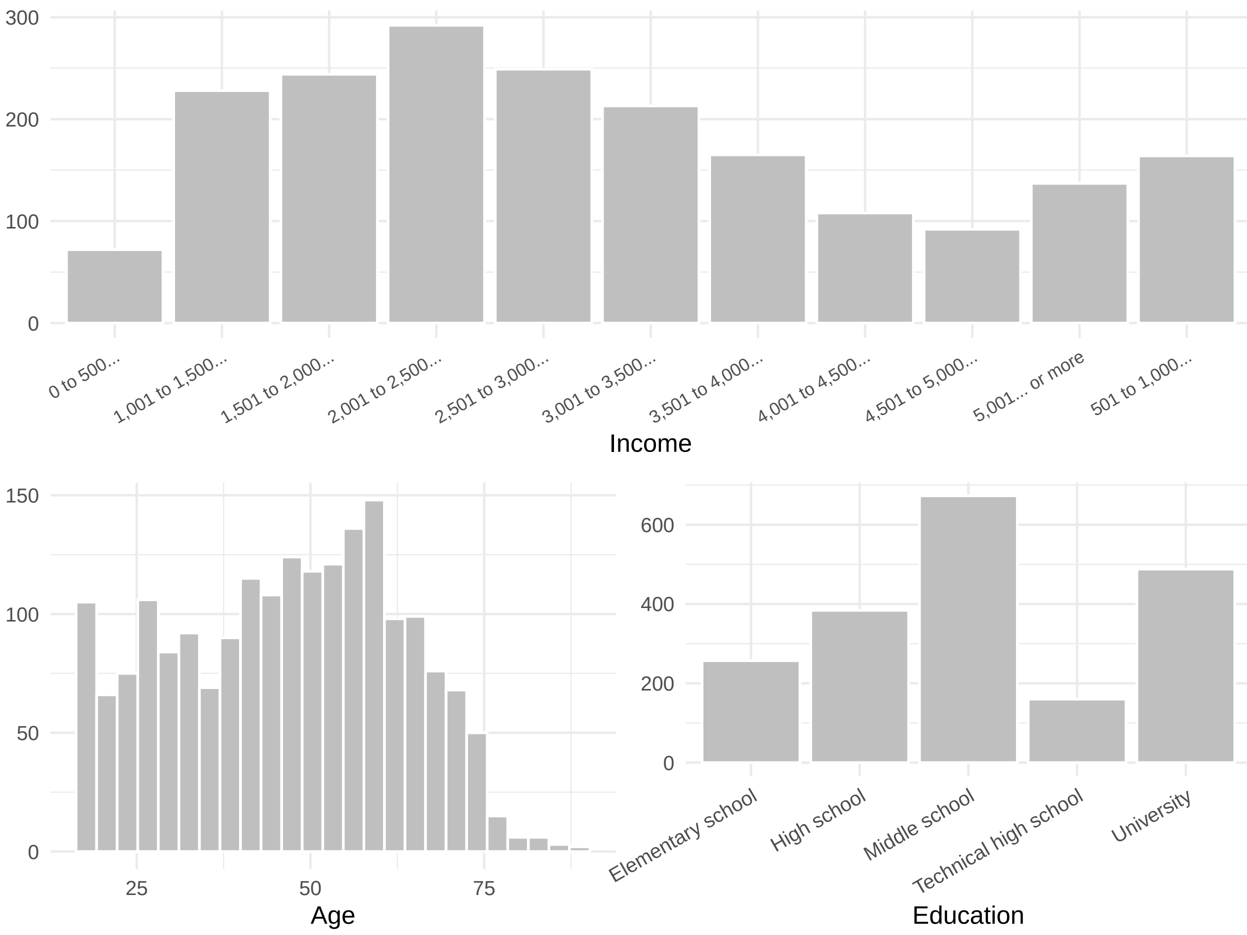

Figure 1: Descriptive distributions of key demographic variables

Remark: Alternatively, we compare the average values of the variables across the control group (given fake source, assume value `0`) and the treatment group (given true source, assume value `1`), shown in Table 2. The results indicate no substantial differences in means between the two groups, suggesting a reasonable degree of balance. The distributions appear broadly similar across most variables, with the exception of `age`, where some differences are observed. In addition, the sample appears to be slightly skewed toward `male` respondents relative to `female` respondents.

Table 2: Mean comparison between treatment and control groups

Group	Belief	Age	Gender	Education	Income
Fake source	3.15	47.15	0.51	3.09	5.74
Real source	3.78	45.89	0.47	3.09	5.69

Following the literature, we decide to include quadratic transformation of `income`, `age` and `education`, as well as consider non-linear relationship between controls and outcomesto be used in analysis section.

2.2 Statistical Modelling and Analysis

Let us recall some necessary notions, let $Y_i(1)$ and $Y_i(0)$ denote the potential outcomes for individual i under exposure to a real source and a fake source, respectively. The treatment indicator is denoted by D_i, where $D_i = 1$ corresponds to the real-source treatment and $D_i = 0$ corresponds to the fake-source control condition.

The main causal estimand is the average treatment effect, that is whether real-source exposure changes belief and sharing behavior on average:

$$ATE = \mathbb{E}[Y_i(1) - Y_i(0)].$$

To study treatment heterogeneity, that is whether the treatment effect differs across demographic profiles, we also consider the conditional average treatment effect:

$$CATE(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x],$$

where X_i denotes observed demographic characteristics such as age, education, gender, and income.

Because the treatment is experimentally assigned, identification relies on the random assignment assumption:

$$(Y_i(1), Y_i(0)) \perp D_i \mid X_i.$$

Under overlap,

$$0 < \mathbb{P}(D_i = 1 \mid X_i = x) < 1,$$

differences in outcomes between treated and control individuals can be interpreted causally. Covariates are included mainly to improve precision and to explore heterogeneous treatment effects rather than to solve selection bias.

The empirical analysis combines *conventional econometric methods* with modern *causal machine learning* techniques. We first estimate average treatment effects using *simple linear regression, multiple linear regression, logistic regression* for binary sharing outcomes, and *LASSO-based* methods. We then examine treatment heterogeneity, focusing mainly on the belief outcome, using *causal trees, T-learners, and Double Machine Learning*. The final summary table compares the estimated conditional average treatment effects across these approaches, highlighting how conventional econometric estimates relate to more flexible machine learning-based methods.

2.2.1 Conventional Econometric methods

Table 3: T-test results for source treatment

Outcome	Estimate	t-statistic	p-value	N
Belief	-0.624	-9.021	0.000	1979
Email sharing	-0.019	-1.830	0.067	1930
Facebook sharing	-0.051	-2.816	0.005	1210
Twitter sharing	-0.043	-1.096	0.274	292
WhatsApp sharing	-0.029	-2.070	0.039	1639

The baseline specification is a simple linear model:

$$Y_i = \alpha + \tau D_i + \varepsilon_i,$$

where Y_i is the outcome of interest and τ captures the average source treatment effect. We then add demographic controls:

$$Y_i = \alpha + \tau D_i + X_i'\beta + \varepsilon_i.$$

To allow for nonlinear relationships between controls and outcomes, we also estimate an extended specification:

$$Y_i = \alpha + \tau D_i + X_i'\beta + Q_i'\gamma + \varepsilon_i,$$

where Q_i includes squared terms for age, education, and income. Throughout the linear models, standard errors are adjusted using the HC1 heteroskedasticity-robust correction. We do not account for serial correlation, as the data do not exhibit any clear clustered or panel structure.

We begin by estimating simple linear regressions and conducting t-tests to examine our hypotheses. These methods allow us to assess whether differences in average belief and sharing intentions between the two groups are statistically significant. The results provide strong support for *H1a*, while the evidence for *H1b* is more mixed.

Remark: As shown in Table 3, the estimates indicate a positive and statistically significant average treatment effect of approximately 0.62 on `belief`. We also observe a positive and significant effect for `Facebook` sharing, suggesting that being in the *treatment* group increases the probability of sharing via `Facebook` by about 5 percentage points. In addition, the t-test for `WhatsApp` sharing shows a significant effect. In contrast, the estimated effect on `email` sharing is not statistically different from zero. For `Twitter`, where the number of observations is relatively limited, no significant treatment effect is detected.

Remark: Since the sharing intention variables are binary, a *logistic regression model* may be more appropriate than a *linear probability model*. We perhaps give a illustration for the comparison between the *linear probability model* and the *logit model* for `Facebook sharing` in Figure 2, which unsurprisingly similar predicted probabilities are shown across the control and treatment groups. Since the coefficient estimates from the *logit model* are less straightforward to interpret, we continue using the *linear probability model* in the main analysis for ease of interpretation.

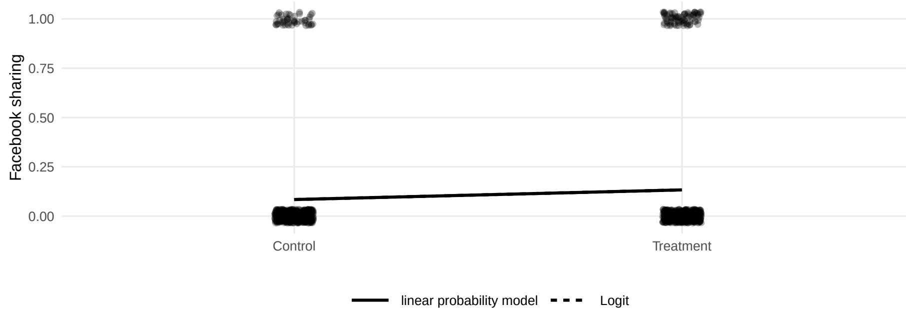

Figure 2: Logit and LPM comparison for Facebook sharing

Table 4: Correlation between treatment and control variables

Variable	Correlation	p-value
Age	-0.039	0.081
Gender	-0.043	0.057
Income	-0.009	0.694
Education	-0.001	0.981

We next examine whether adding control variables changes the estimated treatment effects. The correlation checks in Table 4 suggest only weak associations between the treatment indicator and the main controls, which is consistent with the randomized design. Nevertheless, we estimate multiple linear regression models with demographic controls and then extend the specification by adding quadratic terms for `age`, `education`, and `income`. Standard errors are adjusted using the HC1 heteroskedasticity-robust correction.

2.2.1.1 LASSO regression methods As a final step in the linear econometric model section, we estimate LASSO-based specifications. Standard LASSO solves the penalized least-squares problem

$$\min_{\beta} \left\{ \frac{1}{n} \sum_{i=1}^{n} (Y_i - X_i'\beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}.$$

The penalty term encourages sparse models by shrinking some coefficients exactly to zero. This is useful for variable selection, but it can be problematic for causal inference if the treatment variable or important confounders are removed. For this reason, we also use Double LASSO. The idea is to select controls that predict the outcome and controls that predict the treatment assignment, then estimate the treatment effect using the union of selected controls. This protects the treatment-effect estimate from purely prediction-oriented variable selection.

In this project, LASSO is used mainly as a robustness check against the ordinary linear models. Table 5 summarizes the estimated treatment effects from the simple linear regression, multiple linear regression, Post-LASSO, and Double LASSO specifications.

Remark: According to Table 5, The estimated CATE for `belief` decreases as additional controls are included, suggesting that controlling for observed characteristics leads to more reliable estimates. Moreover, some control variables influence the magnitude of the treatment effect; for instance, the Post-LASSO results suggest that `income` has a positive linear association with `belief.` For `email sharing`, the estimated CATE remains similar across all models. An effect of `gender` is only captured in the MLR1 (multiple linear regression 1) specification, where *quadratic terms* are not included. For `Facebook sharing`, the estimated CATE is consistent across models. A U-shaped nonlinear effect of `education` emerges when quadratic terms (MLR2-multiple linear regression 2) are included, indicating that individuals with either low or high levels of `education` are more likely to share via `Facebook` compared to those with moderate levels of `education`, holding other factors constant.

Table 5: Treatment effect on belief across linear econometric models

	Dependent variable:			
	y	paste(y, "~.")		paste(y, "~d_treatment_source")
	Post-LASSO	MLR2	MLR1	SLR
	(1)	(2)	(3)	(4)
Source treatment	0.607*** (0.068)	0.607*** (0.068)	0.606*** (0.068)	0.624*** (0.069)
Education	0.158*** (0.025)	0.234 (0.155)	0.164*** (0.025)	
Gender	−0.008 (0.069)	−0.004 (0.069)	−0.028 (0.069)	
Age	−0.049*** (0.012)	−0.050*** (0.012)	−0.011*** (0.002)	
Income	0.030** (0.013)	0.079 (0.054)	0.023* (0.013)	
Age squared	0.0004*** (0.0001)	0.0004*** (0.0001)		
Education squared		−0.012 (0.024)		
Income squared		−0.004 (0.004)		
Constant	3.803*** (0.271)	3.592*** (0.369)	3.050*** (0.161)	3.153*** (0.046)
Observations	1,945	1,945	1,945	1,979
R^2	0.091	0.091	0.086	0.040
Adjusted R^2	0.088	0.088	0.084	0.039
Note:				*p<0.1; **p<0.05; ***p<0.01
				HC1 robust standard errors in parentheses.

For `Twitter sharing`, linear regression models show no significant effects. However, the Post-LASSO specification suggests a U-shaped relationship with `income`, where sharing initially decreases with income and then increases at higher `income` levels, on average and controlling for other variables. For `WhatsApp sharing`, the estimated CATE is similar across models, while `age` exhibits a nonlinear relationship with sharing behavior.

Table 6: Double LASSO estimate for belief

Outcome	Estimate	t-statistic	p-value
Belief	0.606	8.9	0.0000

Remark: Overall, the Double LASSO estimate for `belief` in Table 6 is close to the estimates obtained from the linear regression specifications from Table 5 and remains statistically significant. In comparison, the simple linear regression tends to produce larger estimated effects for `belief`, `email sharing`, and `Facebook sharing`, while Double LASSO produces a relatively larger estimate for `WhatsApp sharing`.

2.2.2 Causal Machine Learning methods

Treatment heterogeneity can be examined more flexibly using causal machine learning methods. However, before moving to these methods, we first demonstrate a conventional econometric approach. To examine treatment heterogeneity using a conventional econometric approach, we estimate linear models with *interaction terms.* A simple interaction specification takes the form

$$Y_i = \alpha + \tau D_i + X_i'\beta + D_i X_i'\delta + \varepsilon_i.$$

The interaction coefficient δ captures how the treatment effect changes with observed characteristics. This approach is transparent and easy to interpret, but it requires the researcher to specify the relevant interactions in advance. Specifically, we interact the source treatment with selected demographic variables to examine whether the effect on `belief` differs across subgroups.

Table 7: Heterogeneous treatment effects across subgroups (interaction models)

	Dependent variable:			
	Belief			
	Education	Gender	Income	Age
	(1)	(2)	(3)	(4)
treatment	0.432** (0.188)	0.637*** (0.098)	0.883** (0.378)	1.039*** (0.211)
educationHigh school	0.507*** (0.175)			
educationMiddle school	0.180 (0.155)			
educationTechnical high school	0.346 (0.211)			
educationUniversity	0.641*** (0.162)			
treatment:educationHigh school	0.526** (0.243)			
treatment:educationMiddle school	0.017 (0.221)			
treatment:educationTechnical high school	0.318 (0.306)			
treatment:educationUniversity	0.183 (0.233)			
gendermale		0.006 (0.097)		
treatment:gendermale		−0.038 (0.139)		
income1,001 to 1,500€			−0.192 (0.290)	
income1,501 to 2,000€			−0.137 (0.292)	
income2,001 to 2,500€			−0.043 (0.286)	
income2,501 to 3,000€			0.028 (0.287)	
income3,001 to 3,500€			−0.124 (0.290)	
income3,501 to 4,000€			0.098 (0.302)	
income4,001 to 4,500€			−0.196 (0.326)	
income4,501 to 5,000€			0.131 (0.347)	
income5,001€ or more			−0.067 (0.312)	
income501 to 1,000€			−0.301 (0.308)	
treatment:income1,001 to 1,500€			−0.633 (0.429)	
treatment:income1,501 to 2,000€			−0.488 (0.426)	
treatment:income2,001 to 2,500€			−0.227 (0.418)	
treatment:income2,501 to 3,000€			−0.420 (0.424)	
treatment:income3,001 to 3,500€			−0.041 (0.432)	
treatment:income3,501 to 4,000€			−0.205 (0.447)	
treatment:income4,001 to 4,500€			0.140 (0.479)	
treatment:income4,501 to 5,000€			−0.407 (0.496)	
treatment:income5,001€ or more			0.180 (0.460)	
treatment:income501 to 1,000€			−0.225 (0.447)	
age				−0.010*** (0.003)
treatment:age				−0.009** (0.004)
Constant	2.820*** (0.133)	3.166*** (0.070)	3.250*** (0.255)	3.628*** (0.150)
Observations	1,945	1,945	1,945	1,945
R^2	0.080	0.039	0.056	0.063
Adjusted R^2	0.076	0.038	0.046	0.062
Note:	*p<0.1; **p<0.05; ***p<0.01			
	Interaction terms capture heterogeneous treatment effects.			

Remark: The interaction models from Table 7 suggest some evidence of heterogeneous treatment effects across subgroups. In particular, the treatment effect appears to be larger for individuals with higher levels of education (e.g., `high school`) compared to those with only elementary education. There is also some indication that the effect varies with `age`, with the magnitude of the effect tending to decrease as individuals get older, relative to the average treatment effect. However, this approach relies on specifying interaction terms ex ante, which requires assumptions about the functional form of heterogeneity. This is both restrictive and potentially incomplete, especially in the presence of multiple covariates and possible nonlinear interactions. Moreover, as discussed earlier, testing many interaction terms raises concerns related to multiple testing and limited empirical power.

2.2.2.1 Causal tree method *Causal trees* estimate heterogeneous treatment effects by recursively partitioning the covariate space. Instead of imposing a global linear relationship, the algorithm searches for subgroups in which treatment effects differ. Each *terminal node*, or *leaf*, corresponds to a data-driven subgroup with its own estimated CATE. The main advantage is flexibility: causal trees can capture nonlinearities and interactions without explicitly specifying them. We use the honest estimation framework, where the sample is split into a training subsample used to construct the tree and an estimation subsample used to estimate treatment effects within each leaf. This reduces overfitting and improves the validity of treatment effect estimation. We first examine the causal tree results for `belief`.

```
[1] 2
[1] "CT"
```

Table 8: Leaf-level treatment effects from the causal tree

Leaf	Estimate	Std. Error	t-statistic	p-value
1	-1.800	1.029	-1.750	0.081
2	-1.500	0.945	-1.587	0.113
3	-0.700	1.091	-0.642	0.521
4	-0.600	1.179	-0.509	0.611
5	-0.450	0.945	-0.476	0.634
6	-0.300	0.680	-0.441	0.659
7	0.000	0.996	0.000	1.000
8	0.083	1.076	0.077	0.938
9	0.133	0.514	0.259	0.796
10	0.167	1.150	0.145	0.885
11	0.329	0.694	0.473	0.636
12	0.167	0.813	0.205	0.838
14	0.335	0.213	1.576	0.115
15	0.438	0.185	2.376	0.018
16	0.465	0.416	1.118	0.264
17	1.190	0.710	1.677	0.094
18	0.600	1.179	0.509	0.611
19	0.625	1.114	0.561	0.575
20	0.629	0.451	1.394	0.164
21	0.800	1.179	0.679	0.497
22	0.833	1.076	0.775	0.439
23	0.900	0.945	0.952	0.341
24	0.917	1.076	0.852	0.394
25	0.995	0.451	2.206	0.028
26	0.733	1.029	0.713	0.476
27	1.100	0.616	1.787	0.074
28	1.250	0.833	1.500	0.134

Leaf	Estimate	Std. Error	t-statistic	p-value
29	1.364	0.588	2.319	0.021
30	1.375	0.704	1.952	0.051
31	1.385	0.543	2.552	0.011
32	1.421	0.506	2.806	0.005
33	1.533	0.647	2.369	0.018
34	1.583	0.750	2.112	0.035
35	2.000	1.150	1.739	0.082
36	2.077	0.695	2.987	0.003
37	2.500	0.630	3.968	0.000

Remark: The causal tree partitions the sample into multiple subgroups (*leaves*) as in Table 8, each associated with a distinct estimated treatment effect (CATE). The results show substantial heterogeneity, with treatment effects varying considerably across leaves, indicating that the average treatment effect masks important subgroup differences. However, tree-based estimates can be sensitive to sample splitting, pruning choices, and small leaf sizes. Therefore, rather than interpreting leaf indices directly, these leaves should be understood as data-driven combinations of demographic characteristics (e.g., age, education, income). The variation in estimated effects across leaves suggests that individuals with different profiles respond differently to the source treatment. In particular, the dispersion of CATE estimates across leaves is consistent with earlier findings from interaction models, while providing a more flexible, nonparametric characterization of heterogeneity. This highlights the advantage of the causal tree approach in uncovering complex patterns without imposing a specific functional form.

Table 9: Correlation between causal-tree CATE and demographic variables

Variable	Correlation
Age	0.018
Education	0.159
Gender	0.113
Income	0.476

Remark: The results from Table 9 suggest that, among the individuals for whom the causal tree captures heterogeneous treatment effects, `age` and `education` are negatively correlated with the estimated CATE. This implies that older and more educated individuals may be relatively more skeptical of the news report. The correlation with `gender` is very small, suggesting little difference between men and women in this subgroup. In contrast, `income` is positively correlated with the estimated CATE. Compared with the interaction models, the causal tree gives partly similar evidence for heterogeneity by `age` and `gender`, while also suggesting that `education` and `income` may play a more visible role in shaping heterogeneous treatment effects.

The causal-tree analysis is extended to all outcome variables for consistency. However, the main interpretation of treatment heterogeneity remains focused on `belief`, because the sharing outcomes are binary and less stable across platforms. In addition, causal trees can be sensitive to sample splitting and pruning choices, so their subgroup estimates should be interpreted as exploratory rather than definitive.

2.2.2.2 A Meta-Learner method We consider another approach called *meta-learner*, starting with the *T-learner*. The method estimates treatment heterogeneity by fitting two separate outcome models:

$$\mu_1(x) = \mathbb{E}[Y_i \mid D_i = 1, X_i = x],$$

$$\mu_0(x) = \mathbb{E}[Y_i \mid D_i = 0, X_i = x].$$

The individual-level treatment effect is then estimated as

$$\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x).$$

In this project, we use tree-based learners for both treated and control outcome models. The T-learner is intuitive and easy to implement, but it can be unstable when treatment and control sample sizes are very different. Before estimation, we check the balance between the control and treatment subsamples. This helps assess whether the simple T-learner setup is reasonable, or whether an alternative such as the *X-learner* may be more appropriate.

Table 10: Sample size balance for T-learner estimation

Group	N
Control	995
Treatment	950

The sample sizes are sufficiently close for a basic *T-learner* implementation. We also experimented with pruning and optimal tree selection which not intentionally to be included in this report, but this did not provide a clear improvement for individual-level CATE estimation. Therefore, we proceed with a simple tree-based T-learner using the default complexity parameter as an exploratory specification.

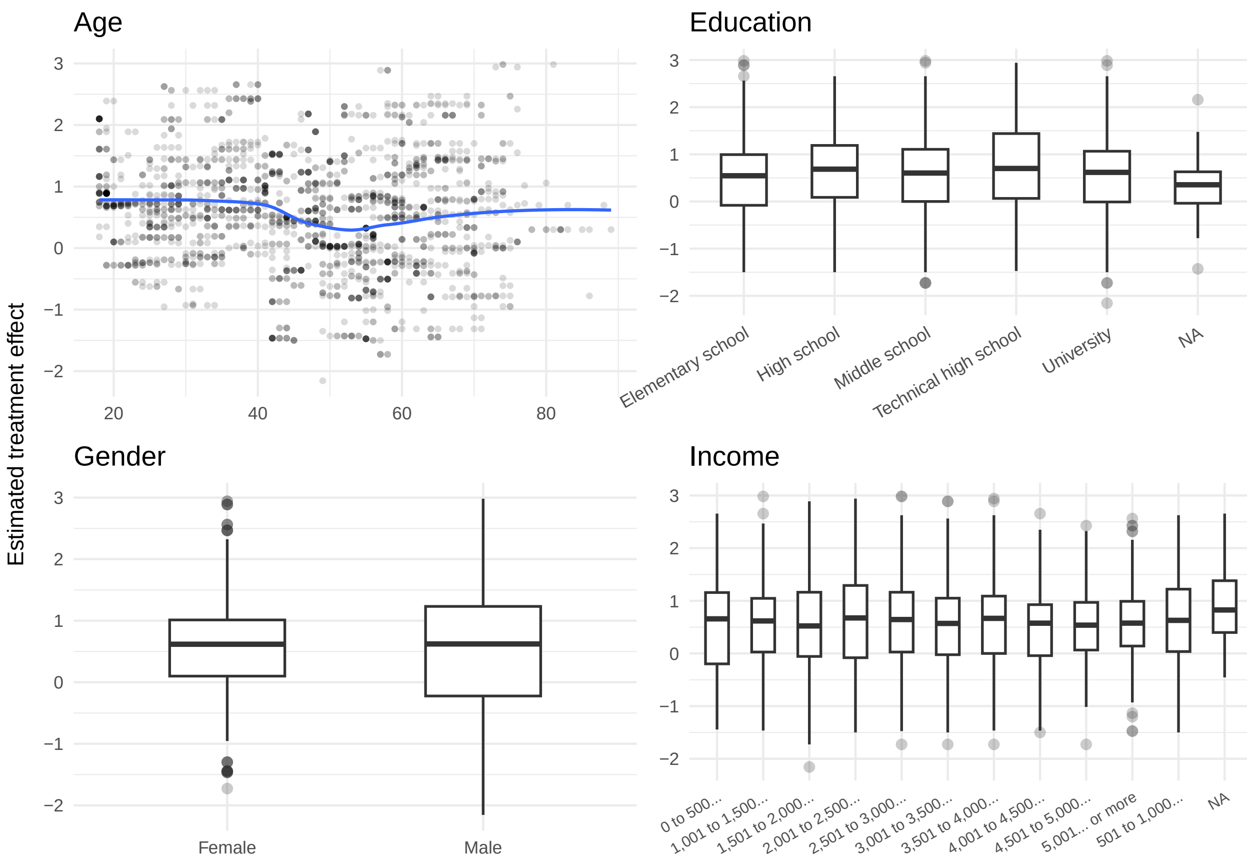

Figure 3: Estimated treatment heterogeneity from the T-learner

Remark: From Figure 3, the smoothed trend for `age`, suggests that the estimated treatment effect is lower among older individuals, although the pattern fluctuates across age groups. This is broadly consistent with the idea that nonlinear age effects may matter, as also reflected in the interaction and quadratic specifications. For `gender`, there is little visible dispersion between the two groups, suggesting limited heterogeneity along this dimension. For `income`, the individual effects do not deviate strongly from the average effect, which is partly consistent with the LMI results. However, some variation across income groups remains, with lower estimated effects around the middle-income range and higher effects among some higher-income groups. For `education`, the estimated effect appears to increase up to a certain level before declining, which differs somewhat from the LMI results. Overall, these patterns should be interpreted cautiously. The T-learner estimates depend on the specific tree configuration and do not provide standard errors for individual-level effects, so the results are best viewed as exploratory rather than definitive.

2.2.2.3 Double Machine Learning method Finally, we apply *Double Machine Learning* as an additional approach to estimate treatment effects. This method flexibly estimate nuisance components while preserving valid treatment-effect estimation through orthogonalization. In a partially linear form, the model can be written as

$$Y_i = \theta D_i + g(X_i) + U_i,$$

$$D_i = m(X_i) + V_i.$$

Here, $g(X_i)$ captures the relationship between controls and the outcome, while $m(X_i)$ captures the relationship between controls and treatment assignment. By residualizing both Y_i and D_i, Double ML reduces sensitivity to first-stage prediction errors. In this project, we use gradient boosting to estimate the nuisance functions. The results are used mainly as a robustness check and as an exploratory tool for treatment heterogeneity.

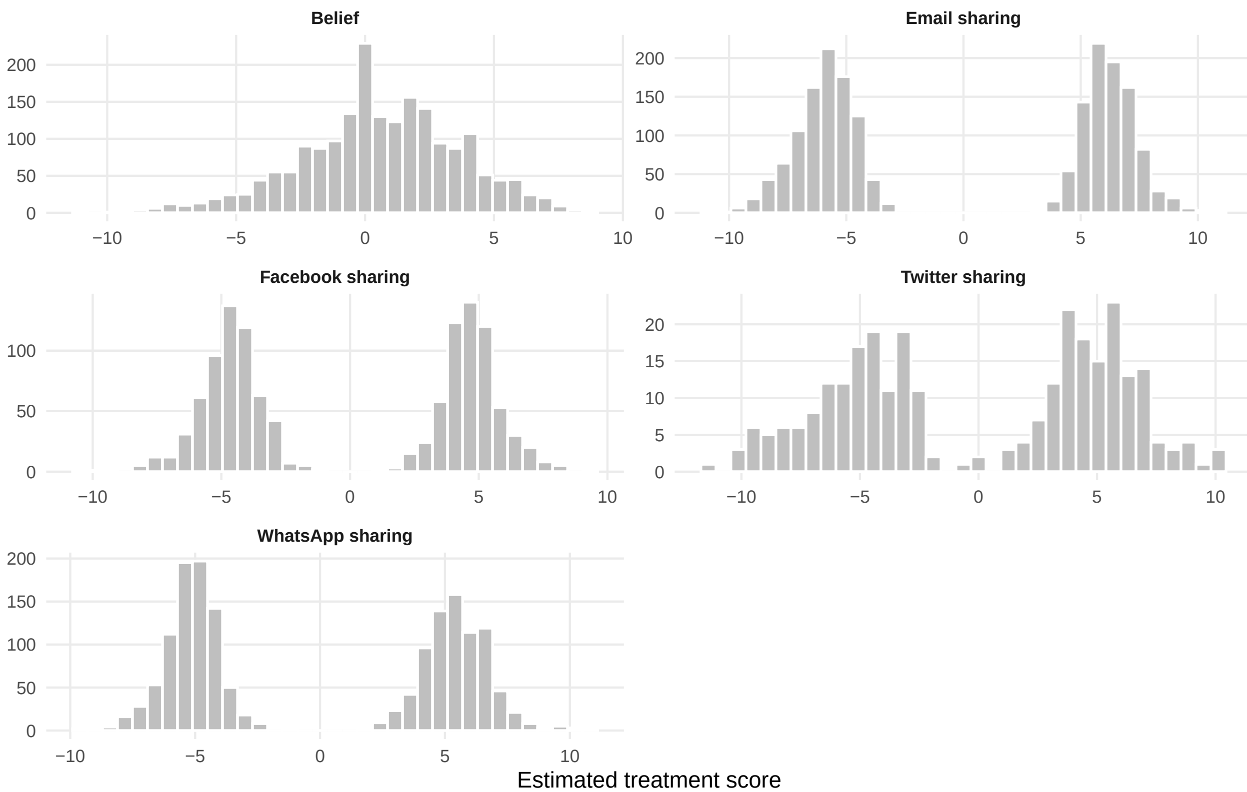

Figure 4: Distribution of Double ML treatment scores

Remark: The Double ML estimates are broadly similar to the simple linear regression estimates, suggesting that the average source effect is reasonably stable across methods. However, the distributions of estimated treatment scores show some variation across outcomes. For some sharing outcomes, the distributions appear more dispersed or bimodal, while the distribution for `belief` is more concentrated. These results should be interpreted cautiously. The estimated individual-level scores may fall outside the natural scale of the original outcomes, and standard errors are not directly available for each individual score. Therefore, the Double ML results are used mainly as an exploratory check on treatment heterogeneity rather than as definitive individual-level causal estimates. This concludes the heterogeneity analysis.

3 Conclusion

The Table 11 summarizes the estimated treatment effects across all methods used in the analysis. For `belief`, all methods produce a positive effect of source credibility. The estimates are broadly similar, although the causal tree produces the largest average leaf-level estimate and Double ML produces the smallest estimate. For `email sharing` and `Twitter sharing`, the estimated effects are generally weak and statistically insignificant across methods. This suggests limited evidence that source credibility affects sharing through these channels. For `Facebook sharing`, the simple regression and Double LASSO estimates suggest a positive effect, while the Double ML estimate is weaker. This indicates that the Facebook result is less robust once more flexible adjustment is introduced. For `WhatsApp sharing`, the simple regression and Double LASSO estimates are positive, while the causal tree and Double ML estimates are weaker or negative. Overall, the evidence for sharing behavior is more mixed than the evidence for belief.

Table 11: Treatment effect estimates across methods

Outcome	Method	Estimate	t-statistic
Belief	SLR / t-test	0.624	9.021
Belief	Double LASSO	0.606	8.904
Belief	Causal tree	0.620	NA
Belief	Double ML	0.602	8.610
Email sharing	SLR / t-test	0.019	1.830
Email sharing	Double LASSO	0.018	1.729
Email sharing	Causal tree	-0.002	NA
Email sharing	Double ML	0.031	0.220
Facebook sharing	SLR / t-test	0.051	2.816
Facebook sharing	Double LASSO	0.049	2.737
Facebook sharing	Causal tree	0.040	NA
Facebook sharing	Double ML	0.044	0.310
Twitter sharing	SLR / t-test	0.043	1.096
Twitter sharing	Double LASSO	0.037	0.960
Twitter sharing	Causal tree	0.041	NA
Twitter sharing	Double ML	0.015	0.044
WhatsApp sharing	SLR / t-test	0.029	2.070
WhatsApp sharing	Double LASSO	0.029	2.106
WhatsApp sharing	Causal tree	-0.002	NA
WhatsApp sharing	Double ML	0.027	0.198

From these results, we revisit our hypotheses. **H1a** is strongly supported: both conventional econometric methods and modern machine learning approaches consistently show that individuals are more likely to believe news from real sources than from fake ones. For **H1b**, the evidence is weaker. The Double ML results suggest no clear causal relationship between exposure to a news source and the willingness to share across channels. In contrast, simple OLS specifications indicate some positive effects, particularly for Facebook and WhatsApp, highlighting that these results are sensitive to the modeling approach.

With the rise of the Internet and social media, the number of information sources has expanded rapidly. A key concern is that unreliable actors can easily imitate legitimate sources. In this project, we examine how source characteristics influence individuals' belief in news reports and their intention to share them. Consistent with our first hypothesis, individuals exhibit stronger belief in reports from real sources. However, the link between belief and sharing behavior is less clear and appears to be channel-dependent. We also find evidence of heterogeneous effects on belief. These differences appear to be driven by nonlinear relationships with demographic characteristics. For example, older individuals tend to be more skeptical, while gender does not show meaningful variation in treatment effects. The results across methods are partly consistent: some patterns (e.g., limited gender heterogeneity) are robust across LMI, causal trees, and T-learners, while others differ depending on the method. For instance, the T-learner suggests an increasing effect with education, whereas the causal tree indicates a negative association. Similarly, for income, LMI and T-learner suggest limited heterogeneity, while the causal tree points to a positive relationship.

Several limitations of our analysis suggest directions for future research. Sharing decisions are influenced by additional factors such as newsworthiness, informational value, emotional content, and framing, which may play a smaller role in belief formation. This may explain why the results for belief and sharing differ. In addition, sharing outcomes are only observed for users of specific platforms, resulting in smaller sample sizes, particularly for Twitter.

Finally, we compare CATE estimates across conventional and machine learning approaches, including Double ML, to assess robustness. While the average effects are relatively stable, the heterogeneity results depend on the chosen method. Overall, our findings have implications for the spread of misinformation. Online platforms play a central role in directing users to content, including potentially misleading sources. Once a source is identified as unreliable, limiting user exposure to such content may help reduce the spread of misinformation.

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